■ MESHMERIZE CHAMPION — ISA LAB REQUISITION MANIFEST (v6.0)

Techfest IIT Bombay | Autonomous Maze Solver | ESP32-S3 Architecture

Streamlined & optimized for maximum scoring performance.

■ REQUISITION LIST (FOR ISA / ROBOTICS LAB INVENTORY)

#	Subsystem	Component	Fallback / Spec	Qty	Why It Matters
1	Main MCU	**ESP32-S3-DevKitC-1-N8**	ESP32-S3-WROOM-1 DevKit	2	Dual-core 240MHz: Core 0 = 1kHz PID, Core 1 = Dijkstra. 1 spare in pit.
2	Motors	**Pololu 20:1 HPCB 12V**	N20 12V 600RPM with Encoder	2	Carbon brushes survive 12.6V (3S) non-stop. Standard motors burn in <5 min.
3	Encoders	**Pololu Magnetic Encoder Pair**	N20 Hall Encoder (12 CPR)	1 pr	480 counts/rev in CHANGE mode → 0.209mm/tick for precise junction detection.
4	Motor Driver	**TI DRV8874 Dual Breakout**	TB6612FNG	1	R_DS(on)=0.12Ω vs TB6612's 1.2Ω — 10× less heat, no brownouts at 3A peak.
5	Line Sensor	**2× Pololu QTR-8RC**	Pololu QTRX-HD-15A	2	16 sensors read via direct GPIO RC-discharge timing. Sub-mm line precision.
6	IMU Sensor	**ICM-42688-P Breakout**	GY-521 MPU-6050 Module	1	Ultra-low noise gyro for accurate 90°/180° pivots. Minimal drift over 3 min.
7	Battery	**Tattu 3S 550mAh 95C LiPo**	GensAce 3S 450mAh 75C	2	52A burst capability — zero voltage sag during motor acceleration.
8	5V Buck	**Pololu D24V22F5 (5V 2.5A)**	MP1584EN module (pre-tuned)	1	Synchronous buck, fixed 5.0V output, no potentiometer drift. Powers MCU+Sensors.
9	3.3V LDO	**TI LP5907 (3.3V 250mA)**	AMS1117-3.3V	1	Ultra-low noise (6.5µV) for clean IMU data acquisition.
10	Wheels	**Pololu 32×7mm Silicone**	Rubber N20 wheels	2	High traction coefficient (µ≈1.0) on arena vinyl floor.
11	Ball Caster	**Pololu 3/8" Stainless Ball**	3/8" plastic ball caster	1	Low-friction 3-point contact.
12	Diode	**SS54 Schottky (5A 40V)**	SS34 (3A)	1	Reverse polarity protection on LiPo input.
13	LED	**Red 5mm High-Brightness**	Standard red LED	1	MANDATORY — Techfest Rule 3. Indicates end-zone hit.
14	Start Switch	**SPST Toggle Switch 3A**	Panel mount SPST	1	MANDATORY — Techfest Rule 2. Master power start.
15	Connectors	**XT30U Connector Pairs**	XT30 Standard	2 pr	High-current battery interface.
16	MCU Socket	**22-Pin 2.54mm Female Headers**	Standard female headers	4 str	Quick MCU swap in pit.
17	Passives	**Decoupling & Snubber Kit**	Assorted SMD/PTH	1 set	Capacitors, voltage divider resistors (27kΩ+100kΩ), snubbers.

■ KEY ELECTRICAL PASSIVES & PIN MAP SUMMARY

Component	Value	Qty	Location	Purpose
Bulk Cap	470µF 16V	2	LiPo / DRV8874 VM	Absorbs motor inrush current spikes.
5V Rail Cap	100µF 10V	1	Pololu Buck Output	Smooths 5V logic supply.
Bypass Caps	100nF 0402	10	Every IC VCC pin	High frequency noise suppression.
Motor Snubbers	100nF 50V	4	Across motor terminals	Eliminates brush arcing EMI.
Batt Divider R1	100kΩ 1%	1	LiPo → GPIO 3 top leg	Voltage divider upper resistor.

Batt Divider R2	**27kΩ 1%**	1	GPIO 3 → GND bottom leg	V_adc_max = 2.68V (Safe for 3.3V ESP32 ADC) ■

■ TECHFEST IIT BOMBAY RULE COMPLIANCE

• **Rule 2 (Master Switch):** SPST toggle switch breaks main LiPo VCC line before buck converter.
• **Rule 3 (End-Zone LED):** Red 5mm LED connected to GPIO 21 activates upon reaching target end zone.
• **Rule 4 (Dimensions):** Chassis layout footprint is 125mm × 110mm ≤ 150mm × 150mm limit.